

IBLOPAY: Gestion des API

Nº	URL	Methode	Description	Paramètres	Retour
1. Module Authentication					

1	/api/auth/login	POST	Authentifie l'utilisateur avec son numero de telephone et son code PIN. Si 2FA est active, retourne requires 2fa:true pour rediriger vers l'écran OTP.	Paramètres (Body JSON) <pre> { "phone_number": "+257XXXXXXXX", "pin": "1234" } </pre>	Succès (pas de 2FA) <pre> { "success": true, "message": "Connexion réussie", "data": { "user": { "user_id": "user-uuid", "first_name": "Jean", "last_name": "Ndayishimiye", "phone_number": "+257XXXXXXXX", "email": "jean@example.com", "cni_number": "1234567890", "photo_url": "https://...", "role_id": "role-uuid", "role": { "role_id": "role-uuid", "name": "ADMIN", "description": "Administrator with management access", </pre>
---	-----------------	------	---	---	--

					<pre> "is_default": true, "created_at": "2026-01-01T00:00:00Z", "updated_at": "2026-01-01T00:00:00Z" }, "status": "ACTIVE", "permissions": ["USER_READ", "DASHBOARD_VIEW"], "created_at": "2026-01-01T00:00:00Z", "updated_at": "2026-01-01T00:00:00Z", "created_by": "user-uuid" }, "tokens": { "access_token": "jwt-access-token", "refresh_token": "jwt-refresh-token", "expires_in": 3600,</pre>
--	--	--	--	--	--

					<pre> { "success": false, "message": "Numéro de téléphone ou PIN incorrect", "status_code": 401, "errors": { "phone_number": ["Ce numéro n'est pas associé à un compte"] } } </pre>
2	/api/auth/verify-2fa	POST	Vérifie le code OTP envoyé par SMS/email pour la double authentification. Si valide, retourne les tokens JWT.	<pre> { "phone_number": "+257XXXXXXXX", "otp_code": "123456" } </pre>	Succès <pre> { "success": true, "message": "Connexion réussie", "data": { "user": { "...": "..."}, "tokens": { "access_token": "jwt-access-token", "refresh_token": "jwt-refresh-token", "expires_in": 3600, "token_type": "Bearer" } } } </pre>

					<pre> } Erreur { "success": false, "message": "Code OTP invalide ou expiré", "status_code": 400, "errors": { "otp_code": ["Le code OTP est invalide ou a expiré"] } } </pre>
3	/api/auth/forgot-password	POST	Envoie un code OTP au numéro de téléphone de l'utilisateur pour réinitialiser son PIN.	<pre> { "phone_number": "+257XXXXXXXX" } </pre>	<pre> Succès(json) { "success": true, "message": "Le code OTP a été envoyé à votre numéro de téléphone", "data": { "otp_sent": true, "expires_in": 300, "phone_number": "+257XXXXXXXX" } } Erreur(json) { "success": false, </pre>

					"message": "Ce numéro de téléphone n'est pas associé à un compte", "status_code": 404 }
4	/api/auth/reset-password	POST	Réinitialise le code PIN après vérification du code OTP	<pre>{ "phone_number": "+257XXXXXXXX", "otp_code": "123456", "new_pin": "5678", "confirm_pin": "5678" }</pre>	Succès(json) <pre>{ "success": true, "message": "Votre PIN a été réinitialisé avec succès" }</pre> Erreur(json) <pre>{ "success": false, "message": "Code OTP invalide ou expiré", "status_code": 400, "errors": { "otp_code": ["Le code OTP est invalide"], "confirm_pin": ["Les codes PIN ne correspondent pas"] } }</pre>
5	/api/auth/refresh	POST	Rafraîchit le token d'accès JWT à l'aide du refresh token.	<pre>{ "refresh_token": "jwt-refresh-token" }</pre>	Succès(json) <pre>{ "success": true, "message": "Token refreshed", }</pre>

			Utilisé automatiquement par l'intercepteur HTTP (AuthInterceptor).		<pre> "data": { "user": { "...": "..."}, "tokens": { "access_token": "new-jwt-access-token", "refresh_token": "new-jwt-refresh-token", "expires_in": 3600, "token_type": "Bearer" } } </pre> Erreur(json) <pre> { "success": false, "message": "Token invalide ou expiré", "status_code": 401 } </pre>
6	/api/auth/logout	POST	Invalide le refresh token côté serveur et déconnecte l'utilisateur.	<pre> { "refresh_token": "jwt-refresh-token" } </pre>	Succès(json) <pre> { "success": true, "message": "Déconnexion réussie" } </pre> Erreur(json) <pre> { </pre>

					<pre>"success": false, "message": "Erreur lors de la déconnexion", "status_code": 500 }</pre>
--	--	--	--	--	---

7	/api/users	GET	Récupère la liste de tous les utilisateurs. Accessible uniquement aux administrateurs.	— — —	Succès(json) <pre> { "success": true, "data": { "users": [{ "user_id": "user-uuid", "first_name": "Jean", "last_name": "Ndayishimiye", "phone_number": "+257XXXXXXXXX", "email": "jean@example.com", "cni_number": "1234567890", "photo_url": "https://...", "role_id": "role-uuid", "role": { "...": "..."}, "status": "ACTIVE", "permissions": ["USER_READ", "DASHBOARD_VIEW"], "created_at": "2026-01-01T00:00:00Z", "updated_at": "2026-01-01T00:00:00Z" }], "pagination": { "page": 1, "limit": 20, "total": 50, </pre>
---	------------	-----	---	-------	--

					<pre> "total_pages": 3 } } } </pre>
8	/api/users/{id}	GET	Récupère les informations d'un utilisateur spécifique.	— — —	Succès(json) <pre> { "success": true, "data": { "user": { "...": "..." } } } </pre>
9	/api/users	POST	Crée un nouvel utilisateur. Accessible uniquement aux administrateurs système.	<pre> { "first_name": "Jean", "last_name": "Ndayishimiye", "phone_number": "+257XXXXXXXX", "email": "jean@example.com", "cni_number": "1234567890", "photo_url": "https://...", "role_id": "role-uuid", "status": "ACTIVE" } </pre>	Succès(JSON) <pre> { "success": true, "message": "Utilisateur créé avec succès", "data": { "user": { "...": "..." } } } </pre> Erreur (doublon json) <pre> { "success": false, "message": "Un utilisateur avec ce numéro de téléphone existe déjà", "status_code": 409 } </pre>

10	/api/users/{id}	PUT	Met à jour les informations d'un utilisateur existant	<pre>{ "first_name": "Jean-Pierre", "email": "jp@example.com", "role_id": "new-role-uuid" }</pre>	Succès(json) <pre>{ "success": true, "message": "Utilisateur mis à jour avec succès", "data": { "user": { "...": "..." } } }</pre>
11	/api/users/{id}/status	PATCH	Modifie le statut d'un utilisateur (ACTIVE, SUSPENDED, FROZEN, CLOSED).	<pre>{ "status": "SUSPENDED" }</pre> Valeur possibles: `ACTIVE`, `SUSPENDED`, `FROZEN`, `CLOSED`	Succès(json) <pre>{ "success": true, "message": "Utilisateur suspendu", "data": { "user": { "...": "..." } } }</pre>
12	/api/users/{id}	DELETE	Supprime définitivement un utilisateur. Accessible uniquement aux administrateurs système.	---	Succes(json) <pre>{ "success": true, "message": "Utilisateur supprimé avec succès" }</pre> Erreur(json) <pre>{ "success": false,</pre>

					"message": "Utilisateur non trouvé", "status_code": 404 }
13	/api/roles	GET	Récupère la liste des rôles disponibles pour l'attribution aux utilisateurs.	— — —	Succès(json) { "success": true, "data": { "roles": [{ "role_id": "role-uuid", "name": "SYSTEM_ADMIN", "description": "System Administrator with full access", "is_default": false, "created_at": "2026-01-01T00:00:00Z", "updated_at": "2026-01-01T00:00:00Z" }] } }
14	/api/permissions	GET	Récupérer la liste de toutes les permissions disponibles	— — —	Succès(json) { "success": true, "data": { "permissions": [

			dans le système.		<pre> { "permission_id": "perm-uuid", "name": "USER_CREATE", "resource": "users", "action": "create", "description": "Create Users", "created_at": "2026-01-01T00:00:00Z" }] } } </pre>
2. Module Agents					
1	/api/agents	GET	Récupère la liste paginée de tous les agents. Filtrable par statut, type, et recherche textuelle	— — — <pre> Filtrer : `SUPER_AGENT`, `AGENT` </pre>	Succès(json) <pre> { "success": true, "data": { "agents": [{ "id": "agent-uuid", "code": "SA-001", "firstName": "Jean", "lastName": "Mukiza", "email": "jean.mukiza@iblopay.com", "phone": "+257 79 900 11 22", "cin": "1-234567-89012", </pre>

					<pre>"cardNumber": "CARTE-IBP-2024-0001", "dateOfBirth": "1980-05-15T00:00:00Z", "address": { "province": "Bujumbura Mairie", "commune": "Mukaza", "zone": "Nyakabiga", "colline": "Nyakabiga", "quartier": "Nyakabiga", "completeAddress": "Nyakabiga, Mukaza, Bujumbura", "latitude": "-3.3860", "longitude": "29.3583" }, "nif": "NIF-123456789", "commerceRegister": "RC-2024-001", "status": "ACTIVE", "type": "SUPER_AGENT", "joinDate": "2024-01-01T00:00:00Z", "createdBy": "admin@iblopay.com", "createdAt": "2024-01-01T00:00:00Z", "updatedAt": "2024-01-01T00:00:00Z", "profileImage": "https://...",</pre>
--	--	--	--	--	---

					<pre> "approvalLetter": "https://..." }], "pagination": { "page": 1, "limit": 10, "total": 10, "total_pages": 1 } } } </pre>
2	/api/agents/{id}	GET	Récupère les informations complètes d'un agent, y compris ses sous-agents, équipements électroniques, et dépôts.	— — —	Succès(json) <pre> { "success": true, "data": { "agent": { "id": "agent-uuid", "code": "SA-001", "firstName": "Jean", "lastName": "Mukiza", "email": "jean.mukiza@iblopay.com", "phone": "+257 79 900 11 22", "cin": "1-234567-89012", "cardNumber": "CARTE-IBP-2024-0001", "dateOfBirth": "1980-05-15T00:00:00Z", "address": { "...": "..." }, </pre>

					<pre> "nif": "NIF-123456789", "commerceRegister": "RC-2024-001", "status": "ACTIVE", "type": "SUPER_AGENT", "joinDate": "2024-01-01T00:00:00Z", "createdBy": "admin@iblopay.com", "createdAt": "2024-01-01T00:00:00Z", "updatedAt": "2024-01-01T00:00:00Z", "profileImage": "https://...", "approvalLetter": "https://..." } } } </pre> Erreur(json) <pre> { "success": false, "message": "Agent non trouvé", "status_code": 404 } </pre>
3	/api/agents	POST	Crée un nouvel agent (Super Agent, Agent, etc.).	<pre> { "firstName": "Jean", "lastName": "Mukiza", "email": "jean.mukiza@iblopay.com", </pre>	Succès(json) <pre> { "success": true, </pre>

				<pre> "phone": "+257 79 900 11 22", "cin": "1-234567-89012", "cardNumber": "CARTE-IBP-2024-0001", "dateOfBirth": "1980-05-15", "address": { "province": "Bujumbura Mairie", "commune": "Mukaza", "zone": "Nyakabiga", "colline": "Nyakabiga", "quartier": "Nyakabiga", "completeAddress": "Nyakabiga, Mukaza, Bujumbura" }, "nif": "NIF-123456789", "commerceRegister": "RC-2024-001", "type": "SUPER_AGENT", "password": "motdepasse123", "profileImage": "base64...", "approvalLetter": "base64..." } </pre>	<pre> "message": "Agent créé avec succès", "data": { "agent": { "...": "..." } } </pre>
4	/api/agents/{agent_id)	PUT	Modifier un agent	<pre> { "firstName": "Jean-Pierre", "email": "jp.mukiza@iblopay.com", "status": "ACTIVE" } </pre>	<pre> { "success": true, "message": "Agent mis à jour avec succès", "data": { "agent": { "...": "..." } } } </pre>

5	/api/agents/{id}	DELETE	Supprimer un agent	— — —	{ "success": true, "message": "Agent supprimé avec succès" }
6	/api/agents/stats		Statistiques des agents. Récupère les statistiques globales (KPI cards) pour le tableau de bord des agents	— — —	{ "success": true, "data": { "total": 10, "active": 6, "pending": 2, "blocked": 1, "suspended": 1, "inactive": 0, "totalElectronics": 8, "totalAgents": 25, "totalDeposits": 30, "totalDepositAmount": 12450000, "totalTransactionAmount": 3200000 } }
7	/api/agents/search	GET	Recherche textuelle d'agents par nom, téléphone,	— — —	{ "success": true, "data": { "agents": [

			code, ou numéro de carte.		<pre> { "id": "agent-uuid", "code": "SA-001", "firstName": "Jean", "lastName": "Mukiza", "phone": "+257 79 900 11 22", "cardNumber": "CARTE-IBP-2024-0001", "status": "ACTIVE", "type": "SUPER_AGENT" }] } } </pre>
8	/api/agents/{id}/sub-agents	GET	Récupère la liste des sous-agents rattachés à un agent (Super Agent).	— — —	<pre> { "success": true, "data": { "subAgents": [{ "id": "sub-agent-uuid", "code": "AG-001", "firstName": "Marie", "lastName": "Niyonzima", "email": "marie.niyonzima@iblopay.com", "phone": "+257 79 900 33 44", "status": "ACTIVE", "joinDate": "2024-02-01T00:00:00Z", </pre>

					<pre>"transactions": [{ "id": "txn-uuid", "type": "DEPOSIT", "amount": 500000, "currency": "BIF", "date": "2024-06-10T00:00:00Z", "status": "COMPLETED", "reference": "TXN-2024-001", "description": "Dépôt initial" }] },], "pagination": { "page": 1, "limit": 20, "total": 5, "total_pages": 1 } }</pre>
9	/api/agents/{id}/electronics	GET	Récupère la liste des équipements électroniques assignés à un agent.	— — —	<pre>{ "success": true, "data": { "electronics": [{ "id": "elec-uuid",</pre>

					<pre>"type": "POS_TERMINAL", "brand": "Ingenico", "model": "Move 5000", "serialNumber": "SN-ING-2024-001", "status": "ACTIVE", "assignedDate": "2024-01-15T00:00:00Z", "lastMaintenance": null, "notes": "Terminal principal", "amountInCirculation": 10000000, "currency": "BIF" }] } }</pre>
10	/api/agents/{id}/deposits	GET	Récupère l'historique des dépôts effectués par les sous-agents d'un agent.	---	<pre>{ "success": true, "data": { "deposits": [{ "id": "dep-uuid", "agentId": "sub-agent-uuid", "agentName": "Marie Niyonzima", "amount": 500000, "currency": "BIF",</pre>

					<pre> "date": "2024-06-10T00:00:00Z", "status": "COMPLETED", "reference": "DEP-2024-001" }], "pagination": { "page": 1, "limit": 20, "total": 5, "total_pages": 1 } } } </pre>
11	/api/agents/{id}/fund	POST	Crée une demande d'approvisionnement pour un agent. Utilisé par la page agent-approvisionnement.	<pre> { "receiptReference": "REC-2024-001", "amount": 500000, "bank": "BANQUE_DE_Credit_de_Bujumbura", "branch": "Agence Centrale", "depositDate": "2024-06-10", "trustAccountVerified": true, "reinforcedVerification": false, "adminNote": "Approbation manuelle requis" } </pre>	<pre> { "success": true, "message": "Demande d'approvisionnement soumise avec succès", "data": { "fundingId": "fund-uuid", "status": "PENDING_APPROVAL", "createdAt": "2024-06-10T14:30:00Z" } } </pre>

12	/api/agents/{id}/block	POST	Bloque ou débloque un agent. L'action est sécurisée par OTP.	<pre>{ "action": "block", "otp_code": "123456" }</pre>	<pre>{ "success": true, "message": "Agent bloqué avec succès", "data": { "agent": { "...": "...", "status": "BLOCKED" } } }</pre>
13	/api/agents/verify-otp	POST	Envoie ou vérifie un code OTP pour les actions sensibles (blocage, approvisionnement).	<pre>{ "action": "send", "phone_number": "+257XXXXXXXX" } ----- { "action": "verify", "phone_number": "+257XXXXXXXX", "otp_code": "123456" }</pre>	<pre>{ "success": true, "message": "Code OTP envoyé avec succès", "data": { "expires_in": 300, "cooldown": 30 } } ----- { "success": true, "message": "Code OTP valide" }</pre>
14	/api/agents/{id}/documents	GET	Récupère la liste des documents associés à un agent (CIN, NIF, etc.).	---	<pre>{ "success": true, "data": { "documents": [{</pre>

					<pre>"id": "doc-uuid", "agentId": "agent-uuid", "type": "CIN", "name": "Carte d'identité nationale", "fileName": "cin_jean_mukiza.pdf", "fileSize": 245760, "mimeType": "application/pdf", "uploadedAt": "2024-01-01T10:00:00Z", "verifiedAt": "2024-01-02T14:00:00Z", "verifiedBy": "admin@iblopay.com", "status": "VERIFIED", "expiryDate": "2030-01-01", "comments": "Document valide" }] } }</pre>												
15	/api/agents/{id}/documents	POST	Télécharge un nouveau document pour un agent.	Paramètres (FormData) <table><tr><th>Nom</th><th>Type</th><th>Requis</th><th>Description</th></tr><tr><td> ----- </td><td> ----- </td><td> ----- </td><td> ----- </td></tr><tr><td> file</td><td> File</td><td> Oui</td><td> Le fichier à télécharger</td></tr></table>	Nom	Type	Requis	Description	-----	-----	-----	-----	file	File	Oui	Le fichier à télécharger	<pre>{ "success": true, "message": "Document téléchargé avec succès", "data": { "document": { "...": "..." } } }</pre>
Nom	Type	Requis	Description														
-----	-----	-----	-----														
file	File	Oui	Le fichier à télécharger														

				type string Oui Type de document (`CIN`, `NIF`, etc.) name string Non Nom du document	} }
16	/api/agents/{id}/documents/{docId}	DELETE	Supprime un document spécifique d'un agent.	— — —	{ "success": true, "message": "Document supprimé avec succès" }
3. Module Commission					
1	/api/commissions	GET	Récupère la liste paginée des commissions avec filtrage et tri.	— — —	{ "success": true, "data": { "items": [{ "commissionId": "COM-0001", "transactionId": "TXN-ID-1000", "transactionReference": "TXN-20240001", "agentId": "AGT-001", "agentName": "Jean-Pierre Hakizimana", "superAgentId": "SA-001", "superAgentName": "Alphonse Niyonzima", "amount": 12500, "rate": 0.75,

					<pre> "commissionType": "AGENT_COMMISSION", "status": "CREDITED", "sourceType": "DEPOSIT", "paymentMode": "AGENT", "createdAt": "2026-07-15T14:30:00.000Z", "creditedAt": "2026-07-18T10:00:00.000Z" }], "pagination": { "page": 1, "pageSize": 15, "total": 200, "totalPages": 14 } } } } </pre>
2	/api/commissions/{id}	GET	Récupère les détails d'une commission spécifique.	— — —	<pre> { "success": true, "data": { "commission": { "commissionId": "COM-0001", "transactionId": "TXN-ID-1000", "transactionReference": "TXN-20240001", "agentId": "AGT-001", </pre>

					<pre> "agentName": "Jean-Pierre Hakizimana", "superAgentId": "SA-001", "superAgentName": "Alphonse Niyonzima", "amount": 12500, "rate": 0.75, "commissionType": "AGENT_COMMISSION", "status": "CREDITED", "sourceType": "DEPOSIT", "paymentMode": "AGENT", "createdAt": "2026-07-15T14:30:00.000Z", "creditedAt": "2026-07-18T10:00:00.000Z" } } } </pre>
3	/api/commissions/kpis	GET	Récupère les cartes KPI pour le tableau de bord des commissions. Le contenu varie selon le rôle (agent, super-agent, admin).		<pre> { "success": true, "data": { "kpis": [{ "label": "Commissions Totales", "value": "12 450 000 BIF", "delta": "+15.8%", "deltaType": "increase", </pre>

					<pre> "icon": "bi-cash-stack", "color": "#3b82f6" }, { "label": "En Attente", "value": "3 200 000 BIF", "delta": "45 transactions", "deltaType": "neutral", "icon": "bi-hourglass-split", "color": "#f97316" }, { "label": "Créditées", "value": "8 750 000 BIF", "delta": "140/200 transactions", "deltaType": "increase", "icon": "bi-check-circle", "color": "#22c55e" }, { "label": "Taux Moyen", "value": "1.05%", "delta": "+0.18%", "deltaType": "increase", "icon": "bi-percent", "color": "#a855f7" }, { "label": "Transactions", "value": "200", </pre>
--	--	--	--	--	---

					<pre> "delta": "+23 vs période préc.", "deltaType": "increase", "icon": "bi-arrow-left-right", "color": "#14b8a6" }] } } </pre>
4	/api/commissions/trend	GET	Récupère les données d'évolution des commissions sur une période (utilisé pour le graphique d'évolution).	— — —	<pre> { "success": true, "data": { "labels": ["2026-07-01", "2026-07-02", "2026-07-03"], "agentValues": [125000, 98000, 150000], "superAgentValues": [45000, 32000, 55000] } } </pre>
5	/api/commissions/breakdown/type	GET	Récupère la répartition des commissions par type d'opération (DEPOSIT, WITHDRAWAL, PAYMENT_NFC, TRANSFER).	— — —	<pre> { "success": true, "data": { "breakdown": [{ "type": "DEPOSIT", "amount": 5200000, "count": 85 }, { "type": "WITHDRAWAL", "amount": 3100000, "count": 52 }, </pre>

			Pour graphique donut.		<pre> { "type": "PAYMENT_NFC", "amount": 2150000, "count": 38 }, { "type": "TRANSFER", "amount": 2000000, "count": 25 }] } </pre>
6	/api/commissions/breakdown/status	GET	Récupère la répartition des commissions par statut (PENDING, CREDITED, FAILED). Pour graphique donut.	— — —	<pre> { "success": true, "data": { "breakdown": [{ "status": "CREDITED", "amount": 8750000, "count": 140 }, { "status": "PENDING", "amount": 3200000, "count": 45 }, { "status": "FAILED", "amount": 500000, "count": 15 }] } } </pre>
7	/api/commissions/leaderboard/agents	GET	Récupère le classement des agents par montant de commissions générées.	— — —	<pre> { "success": true, "data": { "leaderboard": [{ "rank": 1, "agentId": "AGT-001", </pre>

					<pre> "agentName": "Jean-Pierre Hakizimana", "totalCommissions": 450000, "transactionCount": 28, "averageRate": 1.25, "trend": "up", "previousPeriodTotal": 380000 }, { "rank": 2, "agentId": "AGT-005", "agentName": "David Mbonimpa", "totalCommissions": 420000, "transactionCount": 22, "averageRate": 1.10, "trend": "stable", "previousPeriodTotal": 410000 }] } </pre>
8	/api/commissions/leaderboard/super-agents	GET	Récupère le classement des super-agents par commissions générées par leur équipe.	— — —	<pre> { "success": true, "data": { "leaderboard": [{ "rank": 1, "agentId": "SA-001", </pre>

					<pre> "agentName": "Alphonse Niyonzima", "totalCommissions": 1250000, "transactionCount": 72, "averageRate": 1.05, "trend": "up", "previousPeriodTotal": 1100000 }] } } </pre>
9	/api/commissions/ hierarchy	GET	Récupère la structure hiérarchique des commissions (super-agent → agents → transactions).	— — —	<pre> { "success": true, "data": { "hierarchy": [{ "superAgentId": "SA-001", "superAgentName": "Alphonse Niyonzima", "totalCommissions": 1250000, "agents": [{ "agentId": "AGT-001", "agentName": "Jean-Pierre Hakizimana", "totalCommissions": 450000, "transactionCount": 28, "commissions": [{ </pre>

					<pre> "commissionId": "COM-0001", "amount": 12500, "status": "CREDITED", "createdAt": "2026-07-15T14:30:00.000Z" }] }] }] } } </pre>
10	/api/commissions/agents	GET	Récupère la liste des agents pour les menus déroulants de filtre.	— — —	<pre> { "success": true, "data": { "agents": [{ "agentId": "AGT-001", "agentName": "Jean-Pierre Hakizimana" }, { "agentId": "AGT-002", "agentName": "Marie-Claire Ndayishimiye" }] } } </pre>

11	/api/commissions/super-agents	GET	Récupère la liste des super-agents pour les menus déroulants de filtre.	— — —	<pre>{ "success": true, "data": { "superAgents": [{ "superAgentId": "SA-001", "superAgentName": "Alphonse Niyonzima" }, { "superAgentId": "SA-002", "superAgentName": "Emmanuel Rwasana" }] } }</pre>
12	/api/commissions/export	GET	Exporte la liste des commissions aux formats CSV ou PDF.	— — —	Content-Type: text/csv (ou application/pdf) Content-Disposition: attachment; filename="commissions_2026-07-23.csv" Fichier CSV ou PDF contenant les données filtrées.
4. Module Transaction					
/api/trans		GET	Récupère la liste paginée des transactions avec filtrage	— — —	<pre>{ "success": true, "data": { "items": [</pre>

actions 1			(type, statut, mode de paiement, wallet, dates, montants).		<pre>{ "transactionId": "txn-uuid", "reference": "TXN-2026-001", "fromWalletId": "wallet-uuid-1", "toWalletId": "wallet-uuid-2", "amount": 500000, "fee": 1250, "transactionType": "TRANSFER", "status": "COMPLETED", "description": "Transfert vers client", "paymentMode": "MOBILE_APP", "metadata": { "channel": "app" }, "createdAt": "2026-07-15T14:30:00.000Z", "completedAt": "2026-07-15T14:30:02.000Z" }], "pagination": { "page": 1, "pageSize": 20, "total": 450, "totalPages": 23 } }</pre>
--------------	--	--	--	--	--

					<pre> "message": "Transaction not found", "status_code": 404 } </pre>
3	/api/transactions/wallet/{walletId}	GET	Récupère les transactions liées à un wallet (comme expéditeur ou destinataire).	— — —	<pre> { "success": true, "data": { "items": [{ "...": "..." }], "pagination": { "page": 1, "pageSize": 20, "total": 12, "totalPages": 1 } } } </pre>
4	/api/transactions/summary	GET	Récupère un résumé agrégé (total, montant, frais, compteurs par statut et par type) selon les filtres.	— — —	<pre> { "success": true, "data": { "summary": { "totalCount": 450, "totalAmount": 123450000, "totalFees": 310000, "countByStatus": { "PENDING": 45, "COMPLETED": 390, "FAILED": 12, "REVERSED": 3 }, "countByType": { "DEPOSIT": 180, </pre>

					<pre> "WITHDRAWAL": 120, "TRANSFER": 100, "PAYMENT_NFC": 40, "SWEEP": 5, "COMMISSION": 5 } } } } </pre>
5	/api/transactions/{id}/sweep	GET	Récupère le détail « sweep » lié à une transaction de type `SWEEP` ou `SWEEP_INVERSE`.	— — —	<pre> { "success": true, "data": { "sweep": { "sweepId": "sweep-uuid", "transactionId": "txn-uuid", "fromWalletId": "wallet-uuid-1", "toWalletId": "wallet-uuid-2", "amount": 10000000, "reason": "PAYMENT_NFC", "status": "COMPLETED", "parentTransactionId": null, "createdAt": "2026-07-15T14:30:00.000Z", "completedAt": "2026-07-15T14:30:03.000Z", "rolledBackAt": null } } } </pre>

6	/api/transactions/{id}/commission	GET	Récupère le détail de la commission liée à une transaction de type `COMMISSION`.	— — —	<pre> { "success": true, "data": { "commission": { "commissionId": "commission-uuid", "transactionId": "txn-uuid", "agentId": "agent-uuid", "amount": 12500, "rate": 2.50, "commissionType": "AGENT_COMMISSION", "status": "CREDITED", "createdAt": "2026-07-15T14:30:00.000Z", "creditedAt": "2026-07-18T10:00:00.000Z" } } } </pre>
7	/api/dashboard/kpis	GET	Récupère les cartes KPI du hub de transactions (volume, aujourd'hui, succès, échecs, en attente, dépôts à valider,	— — —	<pre> { "success": true, "data": { "kpis": { "volumeTotal": { "label": "Volume total", "value": "1 250 000 000 BIF", "delta": "+5.2%", "deltaType": "increase", "extra": "vs 30 derniers jours", "icon": </pre>

			solde compte fiduciaire, e-money en circulation).		"bi-cash-stack", "sparkline": [120, 145, 132, 160, 155, 180] }, "transactionsToday": { "label": "Transactions aujourd'hui", "value": "3 240", "delta": "+8.1%", "deltaType": "increase", "icon": "bi-arrow-left-right" }, "succeeded": { "label": "Réussies", "value": "3 180", "delta": "98.1%", "deltaType": "neutral", "icon": "bi-check-circle" }, "failed": { "label": "Échouées", "value": "38", "delta": "1.2%", "deltaType": "decrease", "icon": "bi-x-circle" }, "pending": { "label": "En attente", "value": "22", "delta": "validation requise", "deltaType": "neutral", "icon": "bi-hourglass-split" }, "depositsToValidate": { "label": "Dépôts à valider", "value": "14", "delta": "3 à valider", "deltaType": "neutral", "icon": "bi-inbox" }, "trustAccountBalance": { "label": "Solde compte fiduciaire", "value": "450 000 000 BIF", "delta": "+2.1%", "deltaType": "increase", "icon": "bi-bank" },
--	--	--	---	--	---

					<pre> "eMoneyInCirculation": { "label": "E-money en circulation", "value": "312 000 000 BIF", "delta": "+1.4%", "deltaType": "increase", "icon": "bi-coin" } } } } </pre>
8	/api/dashboard/quick-actions	GET	Récupère la liste des actions rapides affichées dans le hub.	— — —	<pre> { "success": true, "data": { "quickActions": [{ "label": "Nouveau dépôt", "subtitle": "Valider un dépôt", "type": "primary", "badge": 14, "action": "new-deposit" }, { "label": "Transfert", "subtitle": "Effectuer un transfert", "type": "outline-green", "action": "transfer" }, { "label": "Exporter", "subtitle": "Exporter l'historique", "type": "plain", "action": "export" }] } } </pre>
9	/api/transactions/table	GET	Récupère les lignes du	— — —	<pre> { "success": true, </pre>

			tableau des transactions du hub (peut réutiliser GET /api/transactions). PAGINÉ.		<pre> "data": { "items": [{ "dateTime": "2026-07-15T14:30:00.000Z", "transactionNo": "TXN-2026-001", "type": "TRANSFER", "typePillColor": "#6366F1", "sender": "Jean Hakizimana", "recipient": "Marie Niyonzima", "amount": "500 000 BIF", "fee": "1 250 BIF", "status": "COMPLETED", "statusPillColor": "#22c55e", "channel": "Mobile App", "location": "Bujumbura", "category": "mobile_money" }], "pagination": { "page": 1, "pageSize": 20, "total": 450, "totalPages": 23 } } </pre>
10	/api/transactions/role/{role}	GET	Récupère les transactions selon le rôle de l'utilisateur	— — —	<pre> { "success": true, "data": { "items": [{ "..." : "..." }], </pre>

			connecté (filtrage d'accès serveur).		<pre>"pagination": { "page": 1, "pageSize": 20, "total": 320, "totalPages": 16 } } }</pre>
11	/api/transactions/category/{category}	GET	Récupère les transactions filtrées par catégorie (all, mobile_money, card, agent_network).	— — —	<pre>{ "success": true, "data": { "items": [{ "...": "..." }], "pagination": { "page": 1, "pageSize": 20, "total": 120, "totalPages": 6 } } }</pre>
12	/api/transactions/{transactionId}/detail	GET	Récupère le détail enrichi d'une transaction pour le panneau de détail du hub (expéditeur, destinataire, canaux, frais, commission, etc.).	— — —	<pre>{ "success": true, "data": { "detail": { "transactionId": "txn-uuid", "status": "COMPLETED", "statusColor": "#22c55e", "timestamp": "2026-07-15T14:30:00.000Z", "typeTag": "Transfert", "typeTagColor": "#6366F1", "amountTotal": "500 000 BIF",</pre>

					<pre> "fee": "1 250 BIF", "commission": "250 BIF", "netAmount": "498 500 BIF", "sender": { "initials": "JH", "name": "Jean Hakizimana", "phone": "+257XXXXXXXXX", "walletId": "wallet-uuid-1", "balanceBefore": "1 000 000 BIF", "balanceAfter": "500 000 BIF", "province": "Bujumbura", "commune": "Mukaza", "agency": "Agence Centre" }, "recipient": { "initials": "MN", "name": "Marie Niyonzima", "phone": "+257XXXXXXXXX", "walletId": "wallet-uuid-2", "balanceBefore": "200 000 BIF", "balanceAfter": "700 000 BIF", "province": "Gitega", "commune": "Gitega", "agency": "Agence Gitega" }, "channel": "Mobile App", "reference": "TXN-2026-001", "ip": "197.210.0.1", "device": "Samsung Galaxy S24", </pre>
--	--	--	--	--	--

					<pre> "location": "Bujumbura", "validatedBy": "Admin", "validatedByAvatar": "https://...", "validatedAt": "2026-07-15T14:31:00.000Z", "attachments": ["https://.../reçu.png"] } } } </pre>
13	/api/transactions/{transactionId}/traceability	GET	Récupère la trace (timeline) d'une transaction pour le composant de traçabilité.	— — —	<pre> { "success": true, "data": { "traceability": { "steps": [{ "icon": "send", "timestamp": "2026-07-15T14:30:00.000Z", "description": "Transaction initiée", "amount": "500 000 BIF", "signedAmount": "+500 000 BIF", "reference": "TXN-2026-001" }, { "icon": "check", "timestamp": "2026-07-15T14:30:02.000Z", </pre>

					<pre> "description": "Transaction complétée", "amount": "500 000 BIF", "signedAmount": "-500 000 BIF", "reference": "TXN-2026-001" }], "initialAmount": "500 000 BIF", "currentAmount": "500 000 BIF", "stepCount": 2, "totalDuration": "2s", "status": "COMPLETED", "statusColor": "#22c55e" } } } </pre>
14	/api/dashboard/top-actors	GET	Récupère le classement des acteurs selon l'onglet (super-agents, agents, clients, merchants).	— — —	<pre> { "success": true, "data": { "topActors": [{ "rank": 1, "name": "SA 01 — Alphonse Niyonzima", "volume": 125000000, "volumeFormatted": "125 000 000 BIF", "barPercent": 100 }, { "rank": 2, "name": "SA 02 — Emmanuel Rwas", "volume": </pre>

					<pre> 98000000, "volumeFormatted": "98 000 000 BIF", "barPercent": 78 }] } }</pre>
15	/api/dashboard/operation-types	GET	Récupère la répartition des types d'opérations (pour le graphique donut).	— — —	<pre> { "success": true, "data": { "operationTypes": [{ "type": "Dépôt", "color": "#22c55e", "count": 180, "percentage": 40 }, { "type": "Retrait", "color": "#ef4444", "count": 120, "percentage": 27 }, { "type": "Paielement NFC", "color": "#3b82f6", "count": 40, "percentage": 9 }, { "type": "Transfert", "color": "#f59e0b", "count": 100, "percentage": 22 }] } }</pre>
16	/api/dashboard/alerts	GET	Récupère la liste des	— — —	<pre> { "success": true,</pre>

			alertes/notifications du parking.		<pre> "data": { "alerts": [{ "icon": "warning", "message": "Échec de paiement NFC sur wallet-123", "link": "/transactions/abc" }, { "icon": "info", "message": "Dépôt de 5 000 000 BIF à valider", "link": "/transactions/def" }] } </pre>
17	/api/dashboard/system-activities	GET	Récupère les activités récentes du système.	— — —	<pre> { "success": true, "data": { "activities": [{ "icon": "shield", "action": "Connexion administrateur", "actor": "admin@iblopay.com", "time": "Il y a 5 minutes" }, { "icon": "gear", "action": "Mise à jour des paramètres", "actor": "system", "time": "Il y a 1 heure" }] } } </pre>
18	/api/dashboard/volume-chart	GET	Récupère les données du	— — —	<pre> { "success": true, </pre>

			graphique de volume (24h ou 7 jours selon la période).		<pre> "data": { "chart": { "labels": ["00:00", "01:00", "02:00"], "values": [120, 95, 80] } } </pre>
19	/api/transactions/export	GET	Export serveur des transactions au format CSV (ou PDF). Recommandé pour les gros volumes, en remplacement de l'export client actuel.	— — — (Content-Type: text/csv (ou application/pdf) Content-Disposition: attachment; filename="transactions_2026-07-23.csv")	
5. Module Cartes					
1	/api/cards	GET	Récupère la liste des cartes. Filtrable par statut, type, et recherche textuelle.	— — —	<pre> { "success": true, "data": { "cards": [{ "cardId": "card-uuid", </pre>

					<pre>"cardUid": "04A1B2C3D4E5F6", "userId": "user-uuid", "walletId": "wallet-uuid", "status": "ACTIVE", "transactionCounter": 42, "activatedAt": "2026-07-15T14:30:00.000Z", "activatedBy": "admin@iblopay.com", "blockedAt": null, "replacedAt": null, "oldCardId": null, "cardType": "PHYSICAL", "maskedPan": "5399 **** **** 1234", "holderName": "Jean Hakizimana", "balance": 500000, "parentCardId": null }], "pagination": { "page": 1, "pageSize": 50, "total": 120, "totalPages": 3 } }</pre>
--	--	--	--	--	---

2	/api/cards/{id}	GET	Récupère une carte spécifique par son ID.	— — — —	<pre>{ "success": true, "data": { "card": { "cardId": "card-uuid", "cardUid": "04A1B2C3D4E5F6", "userId": "user-uuid", "walletId": "wallet-uuid", "status": "ACTIVE", "transactionCounter": 42, "activatedAt": "2026-07-15T14:30:00.000Z", "activatedBy": "admin@iblopay.com", "blockedAt": null, "replacedAt": null, "oldCardId": null, "cardType": "PHYSICAL", "maskedPan": "5399 **** **** 1234", "holderName": "Jean Hakizimana", "balance": 500000, "parentCardId": null } } }</pre> ----- <pre>{ "success": false,</pre>
---	-----------------	-----	---	---------	---

					<pre>"message": "Card not found", "status_code": 404 }</pre>
3	/api/cards	POST	Crée une nouvelle carte (statut initial `NEUTRAL`). Le serveur génère `cardId` et `cardUid`.	<pre>{ "userId": "user-uuid", "walletId": "wallet-uuid", "cardType": "PHYSICAL" }</pre>	<pre>{ "success": true, "message": "Carte créée avec succès", "data": { "card": { "cardId": "card-uuid", "cardUid": "04A1B2C3D4E5F6", "userId": "user-uuid", "walletId": "wallet-uuid", "status": "NEUTRAL", "transactionCounter": 0, "activatedAt": null, "activatedBy": null, "blockedAt": null, "replacedAt": null, "oldCardId": null, "cardType": "PHYSICAL" } } }</pre>
4	/api/cards/{id}/activate	POST	Active une carte provisionnée	<pre>{ "otp_code": "123456" }</pre>	<pre>{ "success": true,</pre>

			(statut passe de `NEUTRAL` à `ACTIVE`).	}	<pre>"message": "Carte activée avec succès", "data": { "card": { "...": "...", "status": "ACTIVE", "activatedAt": "2026-07-23T10:00:00.000Z", "activatedBy": "admin@iblopay.com" } }</pre>
5	/api/cards/{id}/block	POST	Bloque une carte (statut passe à `BLOCKED`). Peut être suivi d'une raison.	<pre>{ "reason": "Cartes perdue ou volée" }</pre>	<pre>{ "success": true, "message": "Carte bloquée avec succès", "data": { "card": { "...": "...", "status": "BLOCKED", "blockedAt": "2026-07-23T10:00:00.000Z" } } }</pre>
6	/api/cards/{id}/close	POST	Ferme définitivement une carte (statut `CLOSED`). __État terminal__ — une carte	<pre>{ "reason": "Clôture du compte" }</pre>	<pre>{ "success": true, "message": "Carte fermée avec succès", "data": { "card": { "...": "...", "status": "CLOSED" } } }</pre>

			fermée ne peut pas être réactivée.		} }
7	/api/cards/{id}/replace	POST	Remplace une carte par une nouvelle. L'ancienne passe à `REPLACED` (avec `replacedAt`), une nouvelle carte `NEUTRAL` est créée avec `oldCardId` pointant vers l'ancienne.	{ "reason": "Carte endommagée" }	{ "success": true, "message": "Carte remplacée avec succès", "data": { "oldCard": { "...": "...", "status": "REPLACED", "replacedAt": "2026-07-23T10:00:00.000Z" }, "newCard": { "cardId": "new-card-uuid", "cardUid": "04F6E5D4C3B2A1", "userId": "user-uuid", "walletId": "wallet-uuid", "status": "NEUTRAL", "transactionCounter": 0, "oldCardId": "card-uuid", "cardType": "PHYSICAL" } } }
8	/api/cards/{id}/transactions	GET	Récupère l'historique des transactions effectuées avec une carte.	— — —	{ "success": true, "data": { "transactions": [{

					<pre> "transactionId": "txn-uuid", "cardId": "card-uuid", "amount": 50000, "currency": "BIF", "status": "COMPLETED", "createdAt": "2026-07-20T14:30:00.000Z" }], "pagination": { "page": 1, "pageSize": 20, "total": 5, "totalPages": 1 } } </pre>
9	/api/cards/stats	GET	Récupère les statistiques globales des cartes pour le tableau (toutes les cartes, cartes principales, cartes secondaires, etc.).	— — —	<pre> { "success": true, "data": { "stats": { "total": 120, "active": 80, "blocked": 15, "suspended": 5, "closed": 8, "replaced": 7, "neutral": 5, "parentCards": 90, "secondaryCards": 30 } } </pre>

					<pre> } } </pre>
10	/api/cards/stock	GET	Récupère les lots de cartes en stock (page `card-stock` — actuellement un stub).	— — —	<pre> { "success": true, "data": { "batches": [{ "batchId": "batch-uuid", "quantity": 500, "receivedAt": "2026-07-01T09:00:00.000Z" }] } } </pre>
11	/api/cards/distributions	GET	Récupère l'historique des distributions de cartes (page `card-distribution` — actuellement un stub).	— — —	<pre> { "success": true, "data": { "distributions": [{ "distributionId": "dist-uuid", "cardId": "card-uuid", "distributedTo": "agent-uuid", "distributedAt": "2026-07-15T14:30:00.000Z" }] } } </pre>

					}
12	/api/cards/activate-scanner	POST	Active une carte à partir d'un QR code scanné (composant `card-qr-scanner`).	{ "cardUid": "04A1B2C3D4E5F6", "otp_code": "123456" }	{ "success": true, "message": "Carte activée via QR code", "data": { "card": { "...": "...", "status": "ACTIVE" } } }
Module					